

RCDEPS Physics Testbed — Radiant Chamber Invariance Monitor v1.0

Hardware experiments to compare your radiant-chamber hypothesis against reference baselines — without constraining the design

1) Goal (Your Terms)

Design and fly instruments that treat space as a radiant chamber and allow for the possibility that propagation characteristics may drift slowly over long times, while keeping standard models only as comparison baselines. No claims are pre-rejected; we build tests that can detect tiny changes if they exist.

2) Minimal Parameterization (for Controls, not dogma)

We encode potential deviations with small, dimensionless knobs the ship can estimate online: • Global drift: $c(t) = c_{\text{std}} \cdot (1 + \epsilon_{\text{global}} + \epsilon_{\text{global}} \cdot t)$ with $|\epsilon_{\text{global}}|, |\epsilon_{\text{global}} \cdot t| \ll 1$ • Directional anisotropy: $c(\hat{e}) = c_{\text{std}} \cdot (1 + \kappa \cdot P_{\text{rad}}(\hat{e} \cdot \mathbf{n}_{\text{test}}))$ ($\hat{e}$ is propagation direction; $\mathbf{n}_{\text{test}}$ is a test axis) • Medium-like index: $n_{\text{eff}} = 1 + v(E, B, \rho_{\text{rad}})$ (allows coupling to local field energy density) Autopilot uses these as calibration parameters; if they fit to ~ 0 , we fly the standard tables.

3) Hardware Suite (Ship-Mountable)

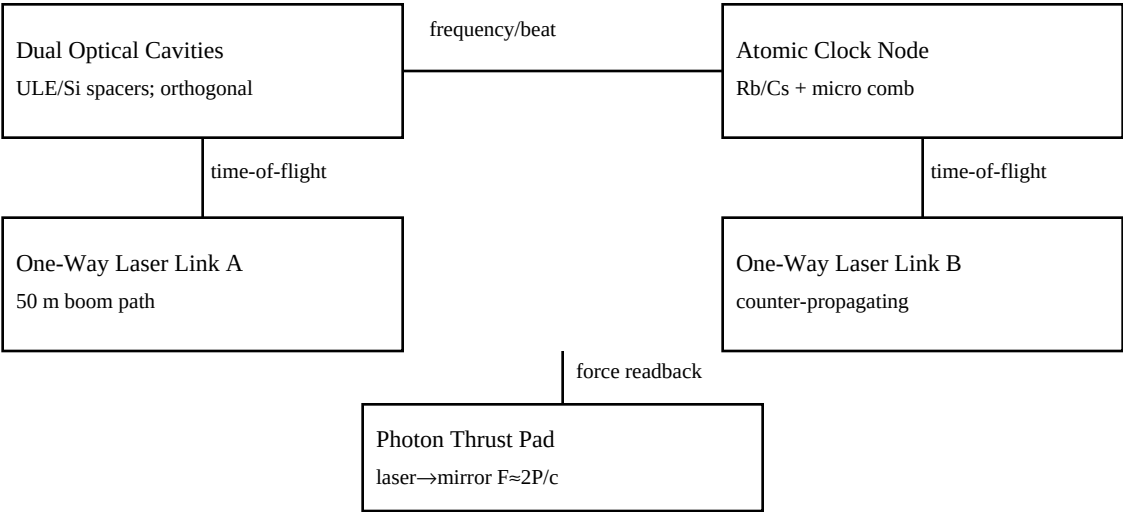


Figure 1. Instrument cluster: orthogonal cavities, atomic clock, two one-way links, and photon-pressure pad.

4) Experiments (Actionable, Ship-Ready)

Exp	What it checks	How it mounts	Acceptance logic
Cavity Beat (orthogonal)	Directional change in propagation (anisotropy) vs. cavity length	Two ultra-stable cavities at 90°, beat the difference	Reluctant to flag thermal/strain → $\kappa \neq 0$; long-term trend
Clock–Cavity Ratio	Drift in α -like combinations vs cavity length	Compare cavity laser to Rb/Cs clock via circulator	Correlated drift across orbits → flag $v(E, B, \rho_{\text{rad}})$
One-Way Links	One-way time-of-flight differences	Two terminals on booms (tens of m), stable fiber beyond transfer	Stable fiber beyond transfer → anisotropy
Photon Pad	Direct measure of radiation-pressure law	Laser on balance-mounted mirror	F vs P deviates from $2P/c$ → v -like effect

5) Controls Integration (No Limits Imposed)

The autopilot treats $\{\epsilon_{\text{■}}, \epsilon_{\text{■}}, \kappa, v\}$ as estimator states. A Kalman filter fuses physics monitors with nav sensors. RCDEPS phase/ β/τ planners read those states to keep thrust timing coherent under any detected propagation shift. If the estimates stay near zero, flight remains standard; if not, guidance adapts without debate.

6) Error Budget & Safeguards

- Temperature/strain: cavities in vacuum cans, multi-stage thermal shielding; reference witness cavity for subtraction.
- Vibration: mount on isochain with reaction-wheel notch filters; record wheel speeds for regression.
- EM fields: μ -metal and Faraday shields; monitor E,B locally to regress $v(E,B,\rho_{\text{rad}})$.
- Data integrity: store raw beat/TOF/force with PPS timestamps; publish null runs beside active runs.

7) Deliverables & Next Steps

Deliverables: instrument mounting drawings, harness list (fiber/DC), estimator state definitions, and acceptance tests. Next: fabricate a 2-cavity breadboard, 10 m lab one-way link, and a photon pad on the thrust stand to validate algorithms before flight.